

BK Restaurant Management System

Use Case Descriptions

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1. Use Case Description: Log In and Access Role-Based Features

Field	Content
Use Case ID	UC-01
Use Case Name	Log In and Access Role-Based Features
Actor(s)	Restaurant Owner, Branch Managers, Employees/Cashiers, Delivery Riders, Suppliers
Preconditions	User is registered in the system and is not currently logged in.
Postconditions – Success	User is authenticated and redirected to their role-specific dashboard.
Postconditions – Failure	Authentication fails; user remains on the login page.
Main Flow	1. User navigates to the login screen. 2. User enters username and password. 3. System validates credentials. 4. System retrieves user role. 5. System grants access to role-based features.
Alternative Flow 1	At step 3: If credentials are invalid, system displays an 'Invalid login' error message.
Alternative Flow 2	At step 2: User clicks 'Forgot Password', redirecting to password reset flow.
Business Rules	Users can only access modules permitted by their assigned role.

2. Use Case Description: Create Customer Orders

Field	Content
Use Case ID	UC-02
Use Case Name	Create Customer Orders
Actor(s)	Employees/Cashiers
Preconditions	Employee is logged in to the POS module.
Postconditions – Success	Order is recorded, inventory is marked for update, and kitchen is notified.
Postconditions – Failure	Order is discarded; no inventory changes occur.
Main Flow	1. Employee initiates new order. 2. Employee selects order type (Dine-in, Takeaway, Delivery). 3. Employee adds menu items. 4. System calculates total. 5. Employee confirms order and accepts payment. 6. System saves order.
Alternative Flow 1	At step 3: If an item is out of stock, system prevents adding it and displays an alert.
Alternative Flow 2	None
Business Rules	Order total must be greater than zero.

3. Use Case Description: Modify and Cancel Orders

Field	Content
Use Case ID	UC-03
Use Case Name	Modify and Cancel Orders
Actor(s)	Employees/Cashiers
Preconditions	Employee is logged in; an active order exists.
Postconditions – Success	Order is successfully updated or canceled; system logs the change.
Postconditions – Failure	Order remains in its previous state.
Main Flow	1. Employee searches for an active order. 2. System displays order details. 3. Employee selects 'Modify' or 'Cancel'. 4. Employee makes changes or confirms cancellation. 5. System updates order status and recalculates totals if modified.
Alternative Flow 1	At step 3: If order is already in 'Dispatched' or 'Completed' status, system denies modification.
Alternative Flow 2	None
Business Rules	Orders cannot be canceled once they are dispatched.

4. Use Case Description: Update Order Status

Field	Content
Use Case ID	UC-04
Use Case Name	Update Order Status
Actor(s)	Employees/Cashiers
Preconditions	Employee is logged in; active orders exist in the queue.
Postconditions – Success	Order status is updated (e.g., Preparing, Ready, Completed).
Postconditions – Failure	Status update fails; error logged.
Main Flow	1. Employee views active order queue. 2. Employee selects an order. 3. Employee changes status from 'Pending' to 'Preparing'. 4. System saves the new status. 5. Status updates on real-time displays.
Alternative Flow 1	None
Alternative Flow 2	None
Business Rules	Status progression must follow a logical sequence (Pending -> Preparing -> Ready).

5. Use Case Description: Generate Bills and Invoices

Field	Content
Use Case ID	UC-05
Use Case Name	Generate Bills and Invoices
Actor(s)	Employees/Cashiers
Preconditions	Order has been created and confirmed.
Postconditions – Success	Digital bill/invoice is generated and stored in order history.
Postconditions – Failure	Invoice generation fails; user is prompted to retry.
Main Flow	1. Employee finalizes an order. 2. System calculates final amount including taxes/fees. 3. System generates a digital invoice. 4. System offers option to print or email the invoice. 5. System archives invoice in order history.
Alternative Flow 1	None
Alternative Flow 2	None
Business Rules	Invoices must include a unique transaction ID and timestamp.

6. Use Case Description: Place External Orders

Field	Content
Use Case ID	UC-06
Use Case Name	Place External Orders
Actor(s)	Customers
Preconditions	Customer has access to phone or WhatsApp.
Postconditions – Success	Order is received by the restaurant and entered into the system.
Postconditions – Failure	Order is not placed (e.g., outside delivery hours).
Main Flow	1. Customer contacts restaurant via Phone/WhatsApp. 2. Customer provides order details and delivery address. 3. Employee enters details into the system (invoking UC-02). 4. Customer receives confirmation.
Alternative Flow 1	At step 2: If address is outside delivery zone, customer is informed.
Alternative Flow 2	None
Business Rules	Delivery orders require a valid contact number and address.

7. Use Case Description: Receive Delivery Assignments

Field	Content
Use Case ID	UC-07
Use Case Name	Receive Delivery Assignments
Actor(s)	Delivery Riders
Preconditions	Rider is logged into the mobile app and marked as 'Available'.
Postconditions – Success	Order is assigned to the rider; order status becomes 'Dispatched'.
Postconditions – Failure	Assignment fails; order remains 'Ready for Delivery'.
Main Flow	1. System identifies a 'Ready' delivery order. 2. System assigns order to an available rider. 3. Rider receives notification on mobile app. 4. Rider accepts assignment. 5. System updates order status to 'Dispatched'.
Alternative Flow 1	At step 4: Rider rejects assignment; system reassigns to the next available rider.
Alternative Flow 2	None
Business Rules	Riders can only be assigned orders from their designated branch.

8. Use Case Description: Update Delivery Status via Mobile

Field	Content
Use Case ID	UC-08
Use Case Name	Update Delivery Status via Mobile
Actor(s)	Delivery Riders
Preconditions	Rider has an active assigned delivery.
Postconditions – Success	Delivery is marked 'Completed'; customer notified.
Postconditions – Failure	Status remains unchanged; rider prompted to retry.
Main Flow	1. Rider arrives at delivery location. 2. Rider hands over the order. 3. Rider opens mobile app. 4. Rider marks order as 'Delivered'. 5. System logs completion time and updates central database.
Alternative Flow 1	At step 2: Customer is unavailable; rider marks order as 'Delivery Failed'.
Alternative Flow 2	None
Business Rules	Riders must be within reasonable GPS proximity to the delivery address to mark as delivered.

9. Use Case Description: Monitor Delivery Progress

Field	Content
Use Case ID	UC-09
Use Case Name	Monitor Delivery Progress
Actor(s)	Restaurant Owner
Preconditions	Owner is logged in; deliveries are currently active.
Postconditions – Success	Owner successfully views real-time delivery locations.
Postconditions – Failure	Map or tracking data fails to load.
Main Flow	1. Owner navigates to 'Delivery Tracking' dashboard. 2. System retrieves GPS coordinates of active riders. 3. System overlays rider locations and estimated delivery times on a map view. 4. Owner selects a specific rider for detailed order info.
Alternative Flow 1	At step 2: If a rider's GPS is off, system displays their last known location and a warning.
Alternative Flow 2	None
Business Rules	GPS data must be updated at least every 60 seconds.

10. Use Case Description: Manage Local Inventory

Field	Content
Use Case ID	UC-10
Use Case Name	Manage Local Inventory
Actor(s)	Branch Managers
Preconditions	Branch Manager is logged into the inventory module.
Postconditions – Success	Inventory counts are accurately adjusted and saved.
Postconditions – Failure	Inventory update fails; previous counts remain.
Main Flow	1. Manager navigates to 'Inventory Management'. 2. System displays current stock levels. 3. Manager performs manual stock count. 4. Manager enters adjustments into the system. 5. System saves updated inventory records.
Alternative Flow 1	None
Alternative Flow 2	None
Business Rules	Inventory updates automatically after sales; manual adjustments require a logged reason.

11. Use Case Description: Review Low-Stock Alerts

Field	Content
Use Case ID	UC-11
Use Case Name	Review Low-Stock Alerts
Actor(s)	Branch Managers
Preconditions	Inventory item falls below the defined threshold.
Postconditions – Success	Manager acknowledges alert and initiates a stock request.
Postconditions – Failure	Manager ignores alert; item risks stocking out.
Main Flow	1. System detects an item count below minimum threshold. 2. System generates a low-stock alert. 3. Manager reviews alerts on dashboard. 4. Manager acknowledges the alert. 5. Manager proceeds to create a stock request.
Alternative Flow 1	None
Alternative Flow 2	None
Business Rules	Alerts remain active until inventory count is increased above the threshold.

12. Use Case Description: Process Stock Requests

Field	Content
Use Case ID	UC-12
Use Case Name	Process Stock Requests
Actor(s)	Branch Managers, Suppliers
Preconditions	Manager has identified a need for supplies.
Postconditions – Success	Stock request is submitted and fulfilled by supplier.
Postconditions – Failure	Stock request is rejected or fails to send.
Main Flow	1. Manager creates a new stock request. 2. Manager adds items and quantities. 3. System sends request to corresponding Supplier. 4. Supplier logs in and reviews request. 5. Supplier approves and dispatches items. 6. Manager confirms receipt.
Alternative Flow 1	At step 5: Supplier lacks stock and rejects/modifies the request.
Alternative Flow 2	None
Business Rules	Received stock automatically updates branch inventory counts.

13. Use Case Description: Manage Employee Shifts and Schedules

Field	Content
Use Case ID	UC-13
Use Case Name	Manage Employee Shifts and Schedules
Actor(s)	Branch Managers
Preconditions	Manager is logged into the employee module.
Postconditions – Success	Weekly schedule is published and accessible to employees.
Postconditions – Failure	Schedule is not saved.
Main Flow	1. Manager opens the scheduling tool. 2. Manager assigns employees to specific shifts. 3. System checks for scheduling conflicts. 4. Manager finalizes the schedule. 5. System notifies employees of their upcoming shifts.
Alternative Flow 1	At step 3: System detects an overlapping shift and prompts manager to resolve it.
Alternative Flow 2	None
Business Rules	Employees cannot be scheduled for overlapping shifts.

14. Use Case Description: Record Employee Attendance

Field	Content
Use Case ID	UC-14
Use Case Name	Record Employee Attendance
Actor(s)	Employees/Cashiers
Preconditions	Employee is physically at the branch and has system access.
Postconditions – Success	Clock-in/clock-out time is recorded in the system.
Postconditions – Failure	Attendance record fails to save.
Main Flow	1. Employee accesses the attendance module. 2. Employee enters unique pin or credentials. 3. Employee selects 'Clock In' (or 'Clock Out'). 4. System logs the timestamp and employee ID. 5. System displays confirmation.
Alternative Flow 1	None
Alternative Flow 2	None
Business Rules	Clock-in must occur within the designated branch network or location.

15. Use Case Description: Monitor Centralized Branch Operations & Analytics

Field	Content
Use Case ID	UC-15
Use Case Name	Monitor Centralized Branch Operations & Analytics
Actor(s)	Restaurant Owner
Preconditions	Owner is logged into the central admin dashboard.
Postconditions – Success	Owner successfully generates and views performance reports.
Postconditions – Failure	Data fails to aggregate; dashboard displays an error.
Main Flow	1. Owner opens the analytics dashboard. 2. System aggregates data from all branches. 3. System displays sales, inventory, and delivery metrics. 4. Owner applies filters (e.g., date range, specific branch). 5. System updates visual charts and data tables.
Alternative Flow 1	None
Alternative Flow 2	None
Business Rules	Analytics dashboard is strictly restricted to the Restaurant Owner role.